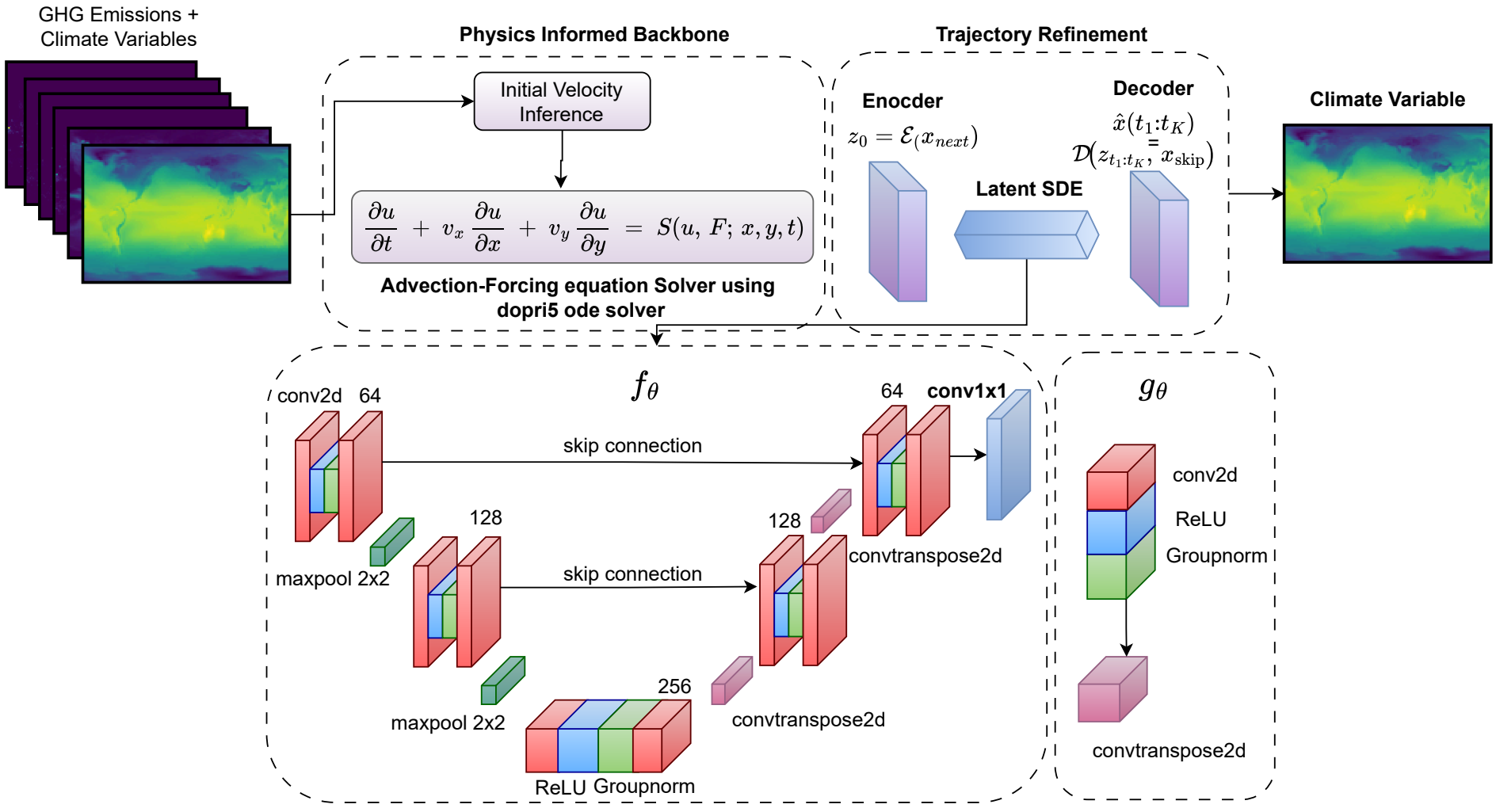

GHG Emissions +
Climate Variables
Physics Informed Backbone
Initial Velocity
Inference
$\frac{\partial u}{\partial t} + v_x \frac{\partial u}{\partial x} + v_y \frac{\partial u}{\partial y} = S(u, F; x, y, t)$
Advection-Forcing equation Solver using
dopri5 ode solver
Trajectory Refinement
Enocder
$z_0 = \mathcal{E}(x_{next})$
Latent SDE
Decoder
$\hat{x}(t_1{:}t_K) = \mathcal{D}(z_{t_1:t_K}, x_{\text{skip}})$
Climate Variable
f_θ
conv2d 64
skip connection
maxpool 2x2
128
skip connection
maxpool 2x2
256
ReLU Groupnorm
128
convtranspose2d
convtranspose2d
64 conv1x1
g_θ
conv2d
ReLU
Groupnorm
convtranspose2d